ADNOC Process Control Specification Alignment

OG RMM Platform — ADNOC-PROC-CTRL-001 Rev 3 Compliance

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Applicable Standard: ADNOC Process Control Specification ADNOC-PROC-CTRL-001 Rev 3
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1. Scope and Purpose

This document demonstrates the OG RMM Platform's alignment with the ADNOC Process Control Specification (ADNOC-PROC-CTRL-001 Rev 3), which governs all SCADA, DCS, and remote monitoring systems deployed across ADNOC Group facilities. It identifies compliant implementations, partial gaps, and the associated Vendor Deviation Requests (VDRs) for any deviations from the standard.

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2. Specification Coverage Summary

Section	Title	Status	VDR Required
4.1	System Architecture Requirements	Compliant	No
4.2	Network Segmentation	Compliant	No
4.3	Historian Configuration	Compliant	No
4.4	Alarm Management (ISA-18.2)	Compliant	No
4.5	Tag Naming Convention	Partial	VDR-001
4.6	Data Retention	Compliant	No
4.7	Cybersecurity (IEC 62443)	Compliant	No
4.8	Functional Safety (IEC 61511)	Compliant	No
4.9	Redundancy and Availability	Compliant	No
4.10	Interface Standards	Partial	VDR-002
5.1	OPC-UA Server Requirements	Compliant	No
5.2	Modbus TCP Requirements	Compliant	No
5.3	DNP3 Requirements	Compliant	No
5.4	MQTT Requirements	Compliant	No
6.1	Reporting Requirements	Compliant	No
6.2	KPI Dashboard Requirements	Compliant	No
6.3	Audit Trail Requirements	Compliant	No
7.1	Localisation (Arabic)	Compliant	No
7.2	Timezone (GST/UTC+4)	Compliant	No
7.3	Currency (AED)	Compliant	No

3. Detailed Alignment

Section 4.1 — System Architecture Requirements

ADNOC Requirement: The system shall implement a minimum 4-tier architecture: Enterprise, DMZ, Control, and Field networks with no direct connectivity between Enterprise and Field tiers.

Platform Implementation: The OG RMM Platform implements a 5-tier IEC 62443 zone architecture: Enterprise (Zone 5), IT/OT DMZ (Zone 4), Control (Zone 3), Field (Zone 2), and Safety (Zone 1). The Rust edge agent operates in Zone 2 and communicates exclusively through the Zone 3 stream processor via authenticated Kafka topics, with no direct Enterprise-to-Field connectivity. The Cybersecurity page's zone/conduit matrix provides real-time visibility of all inter-zone data flows.

Status: Compliant.

Section 4.2 — Network Segmentation

ADNOC Requirement: All OT networks shall be physically or logically separated from IT networks using firewalls rated for industrial use. All cross-zone communication shall be logged.

Platform Implementation: The Helm chart network policies enforce strict egress/ingress rules per namespace. The Rust edge agent DMZ namespace allows only outbound connections to the stream processor on port 9092 (Kafka). All cross-zone API calls are logged in the `audit.events` PostgreSQL table with source zone, destination zone, user, timestamp, and payload hash.

Status: Compliant.

Section 4.3 — Historian Configuration

ADNOC Requirement: All process data shall be stored in a certified historian with minimum 3-year raw data retention and 30-year aggregated data retention. The historian shall support OPC-UA HDA.

Platform Implementation: InfluxDB 2.7 with TimescaleDB continuous aggregates provides the historian layer. Raw telemetry is retained for 3 years (configurable to 15 years in the UAE profile), with 1-hour and 1-day aggregates retained for 30 years. The Go API Gateway exposes an OPC-UA HDA-compatible endpoint via the PI Web API compatibility layer (`/piwebapi/streams/{webId}/recorded`). The ADNOC UAE Helm profile sets `dataRetention.rawTelemetry: "3y"` and `dataRetention.aggregated1d: "30y"`.

Status: Compliant.

Section 4.4 — Alarm Management (ISA-18.2)

ADNOC Requirement: The system shall comply with ISA-18.2 alarm management standard. Maximum alarm rate shall not exceed 10 alarms per operator per 10-minute period. Standing alarm limit: 5. Chattering alarm threshold: 3 occurrences per 10 minutes.

Platform Implementation: The Analytics page ISA-18.2 tab displays real-time alarm rate (alarms/operator/hour), standing alarm count, chattering alarm list, and flood event log. The Alarm Manager Go service enforces the ADNOC-specified thresholds via Temporal workflows. The UAE Helm profile configures `alarmManagement.maxAlarmRate: 10`, `standingAlarmLimit: 5`, and `chattingAlarmThreshold: 3`.

Status: Compliant.

Section 4.5 — Tag Naming Convention

ADNOC Requirement: All historian tags shall follow the ADNOC naming convention: `<FACILITY>-<UNIT>-<INSTRUMENT>-<SUFFIX>`, e.g., `ADMA-PROD-FT-001-PV`.

Platform Implementation: The current platform uses a generic `<WELL_ID>-<SENSOR_TYPE>` naming convention (e.g., `PB47-TUBING_PRESSURE`). Full ADNOC tag naming requires a migration of all existing tag names and a tag alias layer in the historian.

Status: Partial. VDR-001 submitted.

Section 4.6 — Data Retention

ADNOC Requirement: Raw process data: 3 years minimum. Aggregated (1-hour): 15 years. Aggregated (1-day): 30 years. Alarm records: 10 years. Financial records: 10 years. Audit logs: 10 years.

Platform Implementation: All retention periods are configurable via Helm values. The UAE profile sets all ADNOC-required retention periods. TimescaleDB retention policies are applied automatically on deployment.

Status: Compliant.

Section 4.7 — Cybersecurity (IEC 62443)

ADNOC Requirement: The system shall achieve IEC 62443 Security Level 2 (SL-2) for all OT zones. Security Level 3 (SL-3) for Safety zones.

Platform Implementation: The Cybersecurity page implements the full IEC 62443 zone/conduit matrix with SL-2 target for Control and Field zones and SL-3 for the Safety zone. The SIS module enforces SIL-⅔ functional requirements per IEC 61511. The IEC 62443 compliance score is displayed as a live KPI on the Cybersecurity dashboard.

Status: Compliant.

Section 4.10 — Interface Standards

ADNOC Requirement: The system shall provide a certified OPC-UA server with ADNOC-standard address space model. Certification by OPC Foundation required before production deployment.

Platform Implementation: The OPC-UA server is implemented in the Rust edge agent using the `opcua` crate. The address space follows the OPC UA for Oil & Gas Part 1 companion specification. However, formal OPC Foundation certification has not yet been obtained.

Status: Partial. VDR-002 submitted.

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4. Vendor Deviation Requests

VDR-001: Tag Naming Convention Migration

Field	Value
VDR Number	VDR-001
Specification Section	4.5
Deviation Description	Current tag naming uses `<WELL_ID>-<SENSOR_TYPE>` format instead of ADNOC `<FACILITY>-<UNIT>-<INSTRUMENT>-<SUFFIX>`
Justification	The platform was designed for multi-operator use with a generic naming convention. A tag alias layer can map existing tags to ADNOC format without data migration.
Risk Assessment	Low — tag alias layer provides transparent translation; no data loss
Proposed Resolution	Implement a tag alias configuration table in PostgreSQL (`historian.tag_aliases`) and a translation middleware in the Go API Gateway. Estimated effort: 2 weeks.
Target Completion	6 weeks from contract award
ADNOC Approval Required	Yes — ADNOC Process Control Engineering

VDR-002: OPC Foundation Certification

Field	Value
VDR Number	VDR-002
Specification Section	4.10
Deviation Description	OPC-UA server implementation not yet OPC Foundation certified
Justification	The OPC-UA server is fully functional and compliant with the OPC UA specification. Formal certification requires a 3–6 month OPC Foundation testing process.
Risk Assessment	Medium — interoperability with third-party OPC-UA clients may have edge cases
Proposed Resolution	Submit for OPC Foundation CTT (Compliance Test Tool) testing within 30 days of contract award. Interim acceptance based on successful interoperability testing with ADNOC-approved OPC-UA clients (Kepware, Matrikon).
Target Completion	6 months from contract award
ADNOC Approval Required	Yes — ADNOC Instrumentation & Control Engineering

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5. ADNOC Vendor Qualification Checklist

Requirement	Status	Evidence Document
IEC 62443 SL-2 compliance	Complete	IEC 62443 Cybersecurity Dashboard; this document § 4.7
IEC 61511 SIL-2 compliance	Complete	IEC 61511 SIL Documentation Package
ISA-18.2 alarm management	Complete	Analytics page ISA-18.2 tab; this document § 4.4
OPC-UA server	Complete (pending cert.)	Rust edge agent; VDR-002
ADNOC tag naming convention	Partial	VDR-001
Arabic language support	Complete	ME-01 i18n implementation
AED currency	Complete	Financial Ledger UAE profile
UAE data residency	Complete	UAE Helm profile; G42 Cloud deployment
NESA IAS-188 SOA	Complete	NESA_IAS188_Statement_of_Applicability.md
Penetration test (CREST)	Pending	Schedule within 30 days of contract award
ADNOC network security review	Pending	Submit with this package
Factory Acceptance Test (FAT)	Pending	Schedule 8 weeks from contract award
Site Acceptance Test (SAT)	Pending	Schedule 16 weeks from contract award